Lead intake and triage with a local AI in n8n: design and an independent test on 40 unseen messages

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Abstract. *This report describes an n8n workflow for small service businesses that reads each incoming message with an AI model running on the business's own computer, extracts the customer's details, decides what should happen to the message and records it in a spreadsheet, alerting the owner whenever a person needs to act. Fixed rules filter what does not need the AI, every AI answer is checked before it is saved, and the final status is set by rules rather than by the model. On an independent test of 40 messages written by a separate author and sealed before the run, 38 were routed correctly (95%), no urgent message was handled automatically, and names, phone numbers and locations were extracted correctly in all 30 cases each.*

1. The problem

A plumber, an electrician or a cleaning company lives on how fast it answers. Messages arrive through a website form, e-mail or text message, mixed with automatic replies, duplicates and spam. Someone has to read each one, copy the name and phone number, decide whether it is urgent and answer the good leads first. Done by hand, that work is slow and error-prone; done by an AI alone, it brings a different risk: the model may invent a phone number or treat an emergency as routine.

2. How it works

The workflow has three stages (Figure 1).

1. Rules before any AI. Messages with no text or no contact are logged as input errors. Automatic replies, blocked senders (such as "no-reply" addresses) and messages that are too short are discarded with a logged reason. A message identical to an earlier one from the same contact is logged as a duplicate. None of these ever reach the AI, which saves time and removes a source of errors.

2. The AI reads, and every answer is checked. A local model (Llama 3.2, 3 billion parameters, through Ollama) reads one message at a time and fills a fixed form: intent, urgency, name, phone, service, location, deadline, budget and a short summary. The sender's contact never goes to the AI. Before anything is saved, code checks the answer: a phone number must appear in the message, and a name must appear in the message or the sender's name; anything that fails is removed and the message goes to a person. Text that would turn into a spreadsheet formula is neutralized, and long fields are trimmed.

3. Status by rule, record and alert. The status is decided by a fixed order of rules, not by the AI: invalid AI output, spam, urgent, complaint, missing service on a new request, or a suspected invention send the message to a person; everything else is

handled automatically. Each message becomes exactly one row in the spreadsheet, and every message that needs a person triggers an alert to the owner. If the AI is down or takes more than 60 seconds, the message is kept with an error status and can be processed again; nothing is lost.

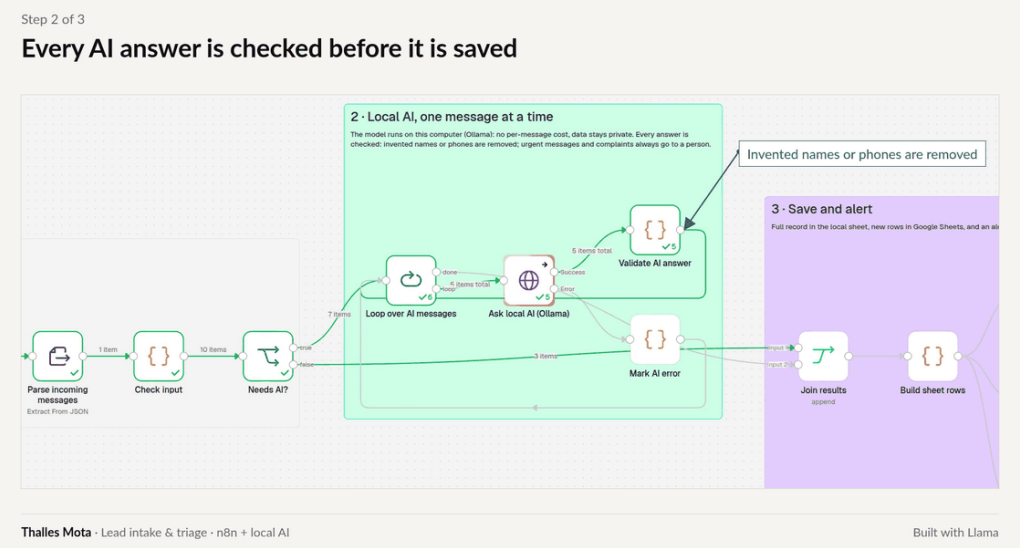


Figure 1. Stage 2 of the workflow in n8n: one message at a time goes to the local AI, and every answer is checked before it is saved.

3. How it was tested

The test was written and reviewed by independent agents that never saw the code, the prompt or any result, and whoever tuned the workflow did not read the messages before the run. The choice of model was recorded before the test, the workflow that ran was checked against its fingerprints, and the test was run once. The 40 messages, in English, cover complete requests and requests missing details, questions, urgent jobs, complaints, spam and scams, automatic replies and other messages the rules filter out, duplicates and one malformed input. Every figure is a count over a count, with a 95% Wilson interval.

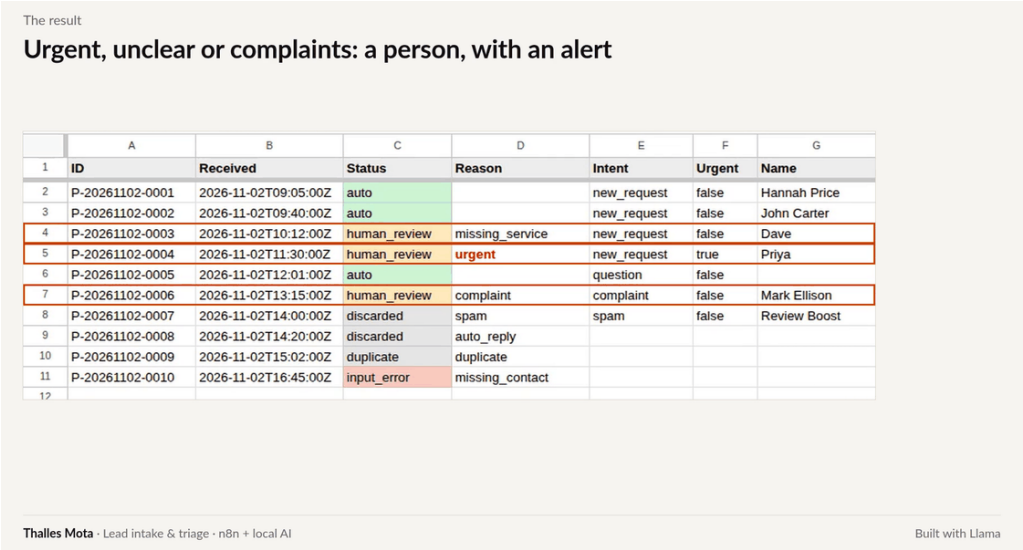
4. Results

Measure	Result
Status correct (automatic, to a person, discarded, duplicate, input error)	38/40 = 95% (83–99%)
Urgent messages handled automatically	0
Requests that should have gone to a person but did not	0
Dangerous formulas written to the spreadsheet	0
Intent classified correctly	33/34
Urgency, name, phone and location extracted correctly	30/30 each
Time for the 40 messages, on a 6 GB laptop GPU	54 s

Table 1. Results of the single test run with the chosen model (Llama 3.2 3B).

Both status errors went the safe way: the message went to a person when it did not need to. A scam message with a deadline ("within 24 hours") was read as a complaint and sent to review instead of being discarded, and a meaningless test message made the AI invent a name, which the check caught and sent to review. Field errors that did not change the status were small: in four cases the deadline was written differently from the text ("next Tuesday" for "next tues"), the service was left empty twice and one budget written as "around 2k" was not extracted.

A larger model did not do better. For information only, Llama 3.1 with 8 billion parameters got 36 of 40 right, with two false urgent flags, one spam message read as "other" and one vague request handled automatically when it should have gone to a person, and it took 145 seconds instead of 54. The smaller model was kept.



	A	B	C	D	E	F	G
1	ID	Received	Status	Reason	Intent	Urgent	Name
2	P-20261102-0001	2026-11-02T09:05:00Z	auto		new_request	false	Hannah Price
3	P-20261102-0002	2026-11-02T09:40:00Z	auto		new_request	false	John Carter
4	P-20261102-0003	2026-11-02T10:12:00Z	human_review	missing_service	new_request	false	Dave
5	P-20261102-0004	2026-11-02T11:30:00Z	human_review	urgent	new_request	true	Priya
6	P-20261102-0005	2026-11-02T12:01:00Z	auto		question	false	
7	P-20261102-0006	2026-11-02T13:15:00Z	human_review	complaint	complaint	false	Mark Ellison
8	P-20261102-0007	2026-11-02T14:00:00Z	discarded	spam	spam	false	Review Boost
9	P-20261102-0008	2026-11-02T14:20:00Z	discarded	auto_reply			
10	P-20261102-0009	2026-11-02T15:02:00Z	duplicate	duplicate			
11	P-20261102-0010	2026-11-02T16:45:00Z	input_error	missing_contact			
12							

Figure 2. The resulting spreadsheet: urgent messages, unclear requests and complaints go to a person, with an alert.

5. What this means in practice

- **For the owner:** new leads land in a spreadsheet with the details filled in, and only the messages that need a person reach them, each with an alert. Urgent jobs never wait in the automatic pile.
- **For privacy and cost:** the AI runs on the business's own computer or server, so messages are not sent to an AI company and there is no per-message fee.
- **For a new client:** the numbers hold for this kind of message. The workflow should be measured again on a sample of the client's real messages before it goes live, and the input is swapped for the client's own channel (e-mail inbox, form or text messages).

6. Limitations

The test has 40 messages in English, written by an independent AI agent rather than taken from real customers, so the interval is wide (83–99%). In the demonstration, messages arrive through a local web address and alerts are written to a file; in a real deployment

they come from the client's inbox or form and alerts go by e-mail or Slack. The workflow does not reply to customers by itself.

7. Authorship and access

Requirements, design, build and evaluation by Thalles Paul Leandro Mota. 26 automated tests cover the rules. The code is not public; a live demonstration is available on request. Built with n8n, Ollama and Llama 3.2. Built with Llama.